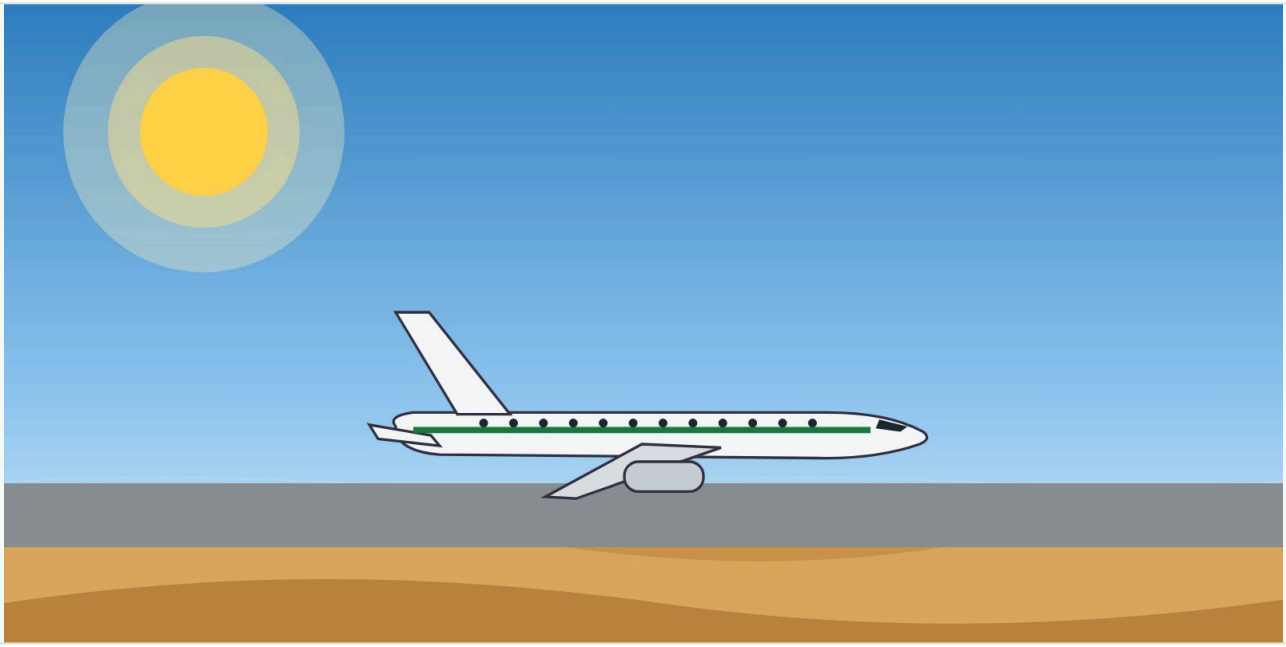
MODULE 1 · PERFORMANCE AND AIRCRAFT SYSTEMS

# Lesson 02: Engines, APU and Aircraft Systems in Hot Weather

Lesson 2 of 6 · about 8 min

**LEARNING OBJECTIVES**

- Apply engine and APU limitations in high temperatures.
- Protect brakes, tyres and avionics in hot conditions.
- Use the fleet-specific hot weather procedures.



Engines, APU, brakes and avionics all have tighter margins in the heat.

## Engines and APU

- Exhaust gas / inter-turbine temperature margins are smaller in hot air. Monitor starts closely and know the **start limits** and the hot start and hung start procedures.
- Keep engine running time on the ground to a minimum, and avoid using high thrust on the ground for long periods.
- APU limits (EGT, load) are also closer in hot weather. Ground power and conditioned air reduce APU use.

## Brakes and tyres

- Brake temperatures rise faster and cool more slowly in hot weather. Respect the minimum **brake cooling times** between flights.
- Avoid riding the brakes during taxi — keep taxi speeds moderate (Lesson 5).
- Tyre pressures increase with temperature; tyres are also stressed by high ground speeds.

## Cockpit, cabin and avionics

- Use ground conditioned air or the APU to cool the cabin and avionics before boarding.
- Watch for avionics overheat cautions and equipment cooling failures.
- Protect crew performance: drink water, manage fatigue and heat stress during long ground times.

## Fleet-specific procedures

| Topic | Boeing 737-800 | Dash 8-Q400 / Q200 |
|---|---|---|
| Hot weather procedures | FCOM Supplementary Procedures — Adverse Weather, hot weather operation (SP 16.17) | AFM / AOM hot weather and high-temperature limitations |
| Engine and APU start limits | FCOM limitations; APU start on battery (SP 7.7) | AFM engine start and ITT limits; APU / ground power as installed |
| Bleeds-off operation | Bleeds-off take-off and landing procedure and performance credit (FCOM / QRH performance) | Bleed and pack settings for take-off per AFM performance charts |
| Fuel system checks | Spar fuel shut-off valve check (SP 12-01) | AOM fuel system procedures |
| Performance | QRH performance inflight / dispatch tables; Jetplan | AFM performance charts or company performance software; Jetplan |

**PREPARE**

Before the summer recurrent training, review the hot weather supplementary procedures and limitations for your type. The values and the exact procedure always come from your aircraft manuals.

**KEY TAKEAWAYS**

- Smaller temperature margins on engine and APU starts — know the limits.
- Respect brake cooling times; avoid riding the brakes.
- Cool the aircraft with ground air or APU; watch avionics cooling.
- Use your type's hot weather procedures (B737 FCOM SP 16.17; Dash 8 AFM/AOM).

**KNOWLEDGE CHECK**

**What is a key consideration for brakes in hot weather?**

A. Brakes need no cooling in summer

B. Brake temperatures rise faster and cool more slowly — respect cooling times

C. Use maximum braking on every landing

D. Set the parking brake during taxi

**Answer: B.** Hot air reduces brake cooling; respect the cooling times between flights.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.